Statistics HW 5-6

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Solution 1:

(i) The Pearson χ^2 test statistic is defined as:

$$T(X) = \sum_{i=1}^{r} \frac{(\nu_i - np_i)^2}{np_i} = \sum_{i=1}^{r} \xi_i^2$$

From this definition, we can see that each ξ_i is defined as:

$$\xi_i = \sqrt{\frac{(\nu_i - np_i)^2}{np_i}} = \frac{\nu_i - np_i}{\sqrt{np_i}}$$

This is the square root of the ratio of the squared difference between the observed frequency ν_i and the expected frequency np_i, divided by the expected frequency np_i.

(ii) To show that the vector $\xi = (\xi_1, \ldots, \xi_r)$ is degenerate, meaning there exists a vector b such that $\xi^\top b = 0$, let's consider the properties of the quantities ξ_i and their relationships.

Definition of ξ_i:

$$\xi_i = \frac{\nu_i - np_i}{\sqrt{np_i}}$$

where ν_i is the observed count in interval Δ_i, and p_i is the probability of a sample falling into Δ_i under the null hypothesis.

Sum of probabilities and counts: The probabilities under the null hypothesis sum to one:

$$\sum_{i=1}^{r} p_i = 1$$

The total number of observations is n, thus:

$$\sum_{i=1}^{r} \nu_i = n$$

Constructing the vector b: Let's choose b such that $b_i = \sqrt{np_i}$ for $i = 1, \ldots, r$. The choice is significant because it relates directly to the denominator of each ξ_i.

Computing $\xi^\top b$:

$$\xi^\top b = \sum_{i=1}^{r} \xi_i b_i = \sum_{i=1}^{r} \left(\frac{\nu_i - np_i}{\sqrt{np_i}} \right) \sqrt{np_i}$$

This simplifies to:

$$\xi^\top b = \sum_{i=1}^{r} (\nu_i - np_i) = \sum_{i=1}^{r} \nu_i - \sum_{i=1}^{r} np_i$$

Given $\sum_{i=1}^{r} \nu_i = n$ and $\sum_{i=1}^{r} np_i = n$, we see that:

$$\xi^\top b = n - n = 0$$

Interpretation: The result $\xi^\top b = 0$ implies that the vector ξ is degenerate in the sense that it lies in a subspace of $\mathbb{R}^r$ where the components of ξ are not independent of each other; they are constrained by the relationship $\sum_{i=1}^{r} \nu_i = n$. This reflects the fact that any increase in one component ν_i necessarily results in a decrease in one or more of the other components ν_j (for $j \neq i$) to keep the total count n constant.

The choice of b demonstrates the linear dependency among the components of ξ, confirming the degeneracy of ξ.

(iii) To utilize the Central Limit Theorem (CLT) to prove that ξ_i has a normal distribution and to find its parameters, we need to express ξ_i in terms of sums of independent random variables:

Expression of ξ_i:

$$\xi_i = \frac{\nu_i - np_i}{\sqrt{np_i}}$$

We rewrite ν_i as a sum of indicator random variables.

Using Indicator Random Variables:
Define indicator random variables I_{ij} such that:

$$I_{ij} = \begin{cases} 1 & \text{if } x_j \in \Delta_i \\ 0 & \text{otherwise} \end{cases}$$

Then, $\nu_i = \sum_{j=1}^{n} I_{ij}$, where j ranges over all the samples.

Normalization:
Substituting the expression for ν_i into ξ_i and normalizing by $\sqrt{n}$:

$$\xi_i = \frac{\sum_{j=1}^{n} I_{ij} - np_i}{\sqrt{np_i}}$$

Application of CLT:
By the CLT, if I_{ij} are independent and identically distributed (i.i.d.) indicator random variables with mean p_i and variance $p_i(1-p_i)$, then as n approaches infinity, the distribution of $\frac{\sum_{j=1}^{n} I_{ij} - np_i}{\sqrt{np_i(1-p_i)}}$ converges to a standard normal distribution.

Parameters of Normal Distribution:
From the CLT, we know that the mean of ξ_i is 0, and its variance is 1. So, ξ_i follows a normal distribution with **mean 0** and **variance 1**.

Therefore, we've shown that as n becomes large, ξ_i follows a normal distribution with mean 0 and variance 1, satisfying the conditions of the CLT.

(iii) To determine $\mathbb{E}[\xi_i \xi_j]$ for both cases $i = j$ and $i \neq j$, we use the expressions we have for ξ_i and the properties of the indicator variables we defined earlier.

For $i = j$:

Recall that

$$\xi_i = \frac{\nu_i - np_i}{\sqrt{np_i}}$$

.

Squared Form:

$$\xi_i^2 = \left(\frac{\nu_i - np_i}{\sqrt{np_i}} \right)^2$$

Expectation:

$$\mathbb{E}[\xi_i^2] = \mathbb{E}\left[\left(\frac{\nu_i - np_i}{\sqrt{np_i}} \right)^2 \right] = \mathbb{E}\left[\frac{(\nu_i - np_i)^2}{np_i} \right].$$

Given $\nu_i = \sum_{j=1}^{n} I_{ij}$ and I_{ij} are independent Bernoulli random variables with $\mathbb{E}[I_{ij}] = p_i$ and $\mathrm{Var}(I_{ij}) = p_i(1-p_i)$, we have $\mathrm{Var}(\nu_i) = np_i(1-p_i)$. Thus:

$$\mathbb{E}[\xi_i^2] = \frac{\mathrm{Var}(\nu_i)}{np_i} = \frac{np_i(1-p_i)}{np_i} = 1 - p_i.$$

For $i \neq j$:
Here we need to consider the covariance between ξ_i and ξ_j.

Covariance Form:

$$\mathbb{E}[\xi_i \xi_j] = \mathbb{E}\left[\frac{\nu_i - np_i}{\sqrt{np_i}} \cdot \frac{\nu_j - np_j}{\sqrt{np_j}} \right].$$

Using Independence:
Since $\nu_i = \sum_{k=1}^{n} I_{ik}$ and $\nu_j = \sum_{k=1}^{n} I_{jk}$, and assuming intervals Δ_i and Δ_j are disjoint for $i \neq j$, the random variables I_{ik} and I_{jk} are independent (one is zero when the other is one, and vice versa). Thus:

$$\mathbb{E}[\nu_i \nu_j] = \mathbb{E}\left[\sum_{k=1}^{n} I_{ik}\right] \mathbb{E}\left[\sum_{k=1}^{n} I_{jk}\right] = np_i \cdot np_j.$$

However, we should recognize that the above holds only when the intervals do not overlap. If they overlap, then we must consider the case of their overlaps.

Zero Covariance for Independence:
If Δ_i and Δ_j are truly disjoint, then:

$$\mathbb{E}[\xi_i \xi_j] = \frac{\mathbb{E}[(\nu_i - np_i)(\nu_j - np_j)]}{\sqrt{np_i np_j}} = \frac{\mathbb{E}[\nu_i \nu_j] - n^2 p_i p_j}{\sqrt{np_i np_j}} = 0.$$

Conclusion:
For $i = j$, $\mathbb{E}[\xi_i \xi_j] = 1 - p_i$.

For $i \neq j$, $\mathbb{E}[\xi_i \xi_j] = 0$, assuming disjoint intervals.

This highlights the independence between ξ_i and ξ_j when $i \neq j$, and shows the variance of ξ_i when $i = j$.

Solution 2:

To test the hypothesis $H_0 : X \sim U[0, 9]$ versus $H_1 : X \not\sim U[0, 9]$ using the Pearson χ^2 test, we follow these steps:

1. Define the null and alternative hypotheses:
   - H_0: The data follows a uniform distribution $U[0, 9]$.
   - H_1: The data does not follow a uniform distribution $U[0, 9]$.
2. Calculate the expected frequencies under H_0:
   - Since $X \sim U[0, 9]$, each interval from 0 to 9 should have the same probability.
   - The total number of observations is $N = 74 + 92 + 83 + 79 + 80 + 73 + 77 + 75 + 76 + 91 = 800$.
   - The probability for each interval is $\frac{1}{10} = 0.1$.
   - The expected frequency for each interval is $E_i = 0.1 \times 800 = 80$.

3. Calculate the observed frequencies:

- The observed frequencies are given as $O_i = [74, 92, 83, 79, 80, 73, 77, 75, 76, 91]$.

4. Compute the χ^2 statistic**: - The χ^2 statistic is calculated as:

$$\chi^2 = \sum_{i=0}^{9} \frac{(O_i - E_i)^2}{E_i}$$

Using the observed and expected frequencies:

$$\begin{aligned}
\chi^2 &= \frac{(74-80)^2}{80} + \frac{(92-80)^2}{80} + \frac{(83-80)^2}{80} + \frac{(79-80)^2}{80} \\
&\quad + \frac{(80-80)^2}{80} + \frac{(73-80)^2}{80} + \frac{(77-80)^2}{80} + \frac{(75-80)^2}{80} \\
&\quad + \frac{(76-80)^2}{80} + \frac{(91-80)^2}{80} \\
&= \frac{36}{80} + \frac{144}{80} + \frac{9}{80} + \frac{1}{80} + \frac{0}{80} + \frac{49}{80} + \frac{9}{80} + \frac{25}{80} + \frac{16}{80} + \frac{121}{80} \\
&= 0.45 + 1.8 + 0.1125 + 0.0125 + 0 + 0.6125 + 0.1125 + 0.3125 + 0.2 + 1.5125 \\
&= 5.13
\end{aligned}$$

5. Determine the degrees of freedom:

- The degrees of freedom $df = k - 1 = 10 - 1 = 9$.

6. Find the critical value from the χ^2 distribution table for $\alpha = 0.05$ and $df = 9$:

- The critical value $\chi^2_{0.05,9} \approx 16.919$.

7. Compare the χ^2 statistic to the critical value:

- Since $\chi^2 = 5.13 < 16.919$, we fail to reject the null hypothesis H_0.

Conclusion:
At the $\alpha = 0.05$ significance level, there is not enough evidence to reject the null hypothesis. Therefore, we conclude that the data does not provide sufficient evidence to suggest that it does not follow a uniform distribution $U[0, 9]$.

Solution 3:

To test the hypothesis $H_0 : \mu_x = \mu_y$ versus $H_1 : \mu_x \neq \mu_y$ using Student's two-sample t-test with the assumption of equal variances, we follow these steps:
1. State the null and alternative hypotheses:

- $H_0 : \mu_x = \mu_y$
- $H_1 : \mu_x \neq \mu_y$

2. Collect the sample data:

- $X = [-1.75, -0.33, -1.26, 0.32, 1.53, 0.35, -0.96, -0.06, 0.42, -1.08]$
- $Y = [-0.29, 0.09, 1.70, -1.09, -0.44, -0.29, 0.25, -0.54, -1.38, 0.32]$

3. Calculate the sample means:

$$\bar{X} = \frac{-1.75 + (-0.33) + (-1.26) + 0.32 + 1.53 + 0.35 + (-0.96) + (-0.06) + 0.42 + (-1.08)}{10}$$
$$= \frac{-2.82}{10} = -0.282$$
$$\bar{Y} = \frac{-0.29 + 0.09 + 1.70 + (-1.09) + (-0.44) + (-0.29) + 0.25 + (-0.54) + (-1.38) + 0.32}{10}$$
$$= \frac{-1.67}{10} = -0.167$$

4. Calculate the sample variances:

$$s_X^2 = \frac{\sum_{i=1}^{10}(X_i - \bar{X})^2}{n_X - 1} = \frac{(-1.75 + 0.282)^2 + (-0.33 + 0.282)^2 + \cdots + (-1.08 + 0.282)^2}{10 - 1}$$
$$s_X^2 = \frac{(2.1550 + 0.002304 + 0.9565 + 0.3624 + 3.2833 + 0.3994 + 0.4597 + 0.04928 + 0.4928 + 0.6368)}{9}$$
$$\approx \frac{8.7475}{9} \approx 0.9775$$
$$s_Y^2 = \frac{\sum_{i=1}^{10}(Y_i - \bar{Y})^2}{n_Y - 1} = \frac{(-0.29 + 0.167)^2 + (0.09 + 0.167)^2 + \cdots + (0.32 + 0.167)^2}{10 - 1}$$
$$s_Y^2 = \frac{(0.01513 + 0.06605 + 3.4857 + 0.8519 + 0.07453 + 0.01513 + 0.1739 + 0.1391 + 1.4714 + 0.2372)}{9}$$
$$\approx \frac{6.5300}{9} \approx 0.7256$$

5. Calculate the pooled variance: -

$$s_p^2 = \frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2} = \frac{(10 - 1) \cdot 0.9775 + (10 - 1) \cdot 0.7256}{10 + 10 - 2}$$
$$s_p^2 = \frac{9 \cdot 0.9775 + 9 \cdot 0.7256}{18} = \frac{8.7975 + 6.5304}{18} = \frac{15.3279}{18} \approx 0.8516$$

6. Calculate the standard error of the difference between the means:
-

$$SE = \sqrt{s_p^2\left(\frac{1}{n_X} + \frac{1}{n_Y}\right)} = \sqrt{0.8516\left(\frac{1}{10} + \frac{1}{10}\right)} = \sqrt{0.8516 \cdot 0.2} = \sqrt{0.17032} \approx 0.4127$$

7. Calculate the t-statistic: -

$$t = \frac{\bar{X} - \bar{Y}}{SE} = \frac{-0.282 - (-0.167)}{0.4127} = \frac{-0.115}{0.4127} \approx -0.2787$$

8. Determine the degrees of freedom:

$$df = n_X + n_Y - 2 = 10 + 10 - 2 = 18$$

9. Find the critical value from the t-distribution table for $\alpha = 0.05$ and $df = 18$:

- For a two-tailed test, the critical t-value at $\alpha = 0.05$ and $df = 18$ is approximately ± 2.101.

10. Compare the t-statistic to the critical value:

- Since $|t| = 0.2787 < 2.101$, we fail to reject the null hypothesis H_0.

Conclusion: At the $\alpha = 0.05$ significance level, there is not enough evidence to reject the null hypothesis. Therefore, we conclude that the data does not provide sufficient evidence to suggest that the means of X and Y are significantly different.

Solution 4:

To use the Wilcoxon signed-rank test to test the hypothesis $H_0 : \mu_x = \mu_y$ versus $H_1 : \mu_x \neq \mu_y$ at the $\alpha = 0.05$ level, we follow these steps:

1. State the null and alternative hypotheses:

- $H_0 : \mu_x = \mu_y$
- $H_1 : \mu_x \neq \mu_y$

2. Collect the sample data:

- $X = [176, 163, 152, 155, 156, 178, 160, 164, 169, 155, 122, 144]$
- $Y = [168, 215, 172, 200, 191, 197, 183, 174, 176, 155, 115, 163]$

3. Calculate the differences $D_i = X_i - Y_i$:

- $D = [8, -52, -20, -45, -35, -19, -23, -10, -7, 0, 7, -19]$

4. Remove zero differences and compute absolute differences:

- Non-zero differences: $D = [8, -52, -20, -45, -35, -19, -23, -10, -7, 7, -19]$
- Absolute differences: $|D| = [8, 52, 20, 45, 35, 19, 23, 10, 7, 7, 19]$

5. Rank the absolute differences:

- Rank $|D|$: $[2, 11, 5, 10, 9, 4, 7, 3, 1.5, 1.5, 4]$

6. Assign signs to the ranks based on the signs of the original differences:

- Signed ranks: $[2, -11, -5, -10, -9, -4, -7, -3, -1.5, 1.5, -4]$

7. **Calculate the test statistic W:**
   - Sum of positive ranks: $W_+ = 2 + 1.5 = 3.5$
   - Sum of negative ranks: $W_- = 11 + 5 + 10 + 9 + 4 + 7 + 3 + 1.5 + 4 = 54.5$
   - Test statistic $W = \min(W_+, W_-) = \min(3.5, 54.5) = 3.5$
8. **Calculate the mean and variance of W:**
   - $n = 11$ (number of non-zero differences)
   - Mean $\mathbb{E}[W] = \frac{n(n+1)}{4} = \frac{11 \cdot 12}{4} = 33$
   - Variance $\mathbb{V}\text{ar}(W) = \frac{n(n+1)(2n+1)}{24} = \frac{11 \cdot 12 \cdot 23}{24} = 126.5$
9. **Calculate the standardized test statistic T:**
   - $T = \frac{W - \mathbb{E}[W]}{\sqrt{\mathbb{V}\text{ar}(W)}} = \frac{3.5-33}{\sqrt{126.5}} = \frac{3.5-33}{11.25} = \frac{-29.5}{11.25} \approx -2.622$
10. **Determine the critical value from the normal distribution:**
    - For a two-tailed test at $\alpha = 0.05$, the critical value is ± 1.96.
11. **Compare the test statistic to the critical value:**
    - Since $T = -2.622$ is less than -1.96, we reject the null hypothesis H_0.

Conclusion:
At the $\alpha = 0.05$ significance level, we reject the null hypothesis. Therefore, we conclude that there is significant evidence to suggest that the medians of the reaction times to light and sound signals are different.

Solution 5:

To answer the question, we will first address part (i) by finding the critical value c so that the test has size α. Then, in part (ii), we will calculate the power W of the test.

(i) Finding c for Size α

Given that $X_1, \ldots, X_n \sim \mathcal{N}(\theta, 1)$, the test statistic $T(X) = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the sample mean of these observations. Under $H_0 : \theta = \theta_0 = 0$, the distribution of $T(X)$ is normal with mean 0 and variance $\frac{1}{n}$, i.e., $T(X) \sim \mathcal{N}(0, \frac{1}{n})$.

The rejection region for the test is defined as $R = \{X : T(X) > c\}$, where c is chosen such that the probability of Type I error (rejecting H_0 when it is true) is α. Therefore,

$$P(T(X) > c \mid \theta = 0) = \alpha.$$

Since $T(X) \sim \mathcal{N}(0, \frac{1}{n})$, standardizing gives:

$$P\left(Z > \frac{c-0}{\sqrt{\frac{1}{n}}}\right) = \alpha,$$

where Z is a standard normal variable. This simplifies to:

$$P\left(Z > c\sqrt{n}\right) = \alpha.$$

Using the standard normal table or Z-score calculator, we find:

$$c\sqrt{n} = z_{1-\alpha},$$

where $z_{1-\alpha}$ is the critical value from the standard normal distribution. For $\alpha = 0.05$,

$$z_{0.95} \approx 1.645.$$

Thus,

$$c = \frac{z_{0.95}}{\sqrt{n}} = \frac{1.645}{\sqrt{n}}.$$

(ii) Finding Test Power W

The power of the test W is the probability that $T(X)$ falls in the rejection region when $\theta = \theta_1 = 1$. Under $H_1 : \theta = 1$, the distribution of $T(X)$ is $\mathcal{N}(1, \frac{1}{n})$. Thus,

$$\begin{aligned} P(T(X) > c \mid \theta = 1) &= P\left(\frac{\frac{1}{n}\sum_{i=1}^{n} X_i - 1}{\sqrt{\frac{1}{n}}} > \frac{c-1}{\sqrt{\frac{1}{n}}}\right), \\ &= P\left(Z > c\sqrt{n} - \sqrt{n}\right), \\ &= P\left(Z > 1.645 - \sqrt{n}\right), \end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$.

Solution 6:

To address the given hypothesis test involving a normal distribution, we follow the steps for computing the likelihood ratio, its logarithm, and the rejection region. We will use the change of variables to transform the test statistic to follow a standard normal distribution.

(i) Compute the Likelihood Ratio $\Lambda(X)$
Given:

- $X_1, \ldots, X_n \sim \mathcal{N}(\theta, \sigma^2)$

- Hypotheses:

$$H_0 : \theta = \theta_0,$$
$$H_1 : \theta \neq \theta_0.$$

The likelihood function under H_0 is:

$$L(\theta_0; X) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \theta_0)^2}{2\sigma^2}\right).$$

The likelihood function under the general alternative hypothesis H_1 is maximized by the sample mean $\bar{X}$:

$$L(\hat{\theta}; X) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \bar{X})^2}{2\sigma^2}\right),$$

where $\hat{\theta} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

The likelihood ratio $\Lambda(X)$ is given by:

$$\Lambda(X) = \frac{L(\theta_0; X)}{L(\hat{\theta}; X)}.$$

Likelihood Ratio Calculation

$$\Lambda(X) = \frac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i-\theta_0)^2}{2\sigma^2}\right)}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i-\bar{X})^2}{2\sigma^2}\right)}.$$

This simplifies to:

$$\Lambda(X) = \exp\left(-\frac{1}{2\sigma^2}\left[\sum_{i=1}^{n}(X_i - \theta_0)^2 - \sum_{i=1}^{n}(X_i - \bar{X})^2\right]\right).$$

Using the fact that:

$$\sum_{i=1}^{n}(X_i - \theta_0)^2 = \sum_{i=1}^{n}[(X_i - \bar{X}) + (\bar{X} - \theta_0)]^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \theta_0)^2,$$

we have:

$$\sum_{i=1}^{n}(X_i - \theta_0)^2 - \sum_{i=1}^{n}(X_i - \bar{X})^2 = n(\bar{X} - \theta_0)^2.$$

Thus, the likelihood ratio becomes:

$$\Lambda(X) = \exp\left(-\frac{n(\bar{X} - \theta_0)^2}{2\sigma^2}\right).$$

(ii) Compute $\lambda(X) = 2\log\Lambda(X)$

$$\lambda(X) = 2\log\Lambda(X) = 2\log\left(\exp\left(-\frac{n(\bar{X}-\theta_0)^2}{2\sigma^2}\right)\right) = -\frac{n(\bar{X}-\theta_0)^2}{\sigma^2}.$$

Distribution of $\lambda(X)$

Under H_0 ($\theta = \theta_0$):

$$\bar{X} \sim \mathcal{N}\left(\theta_0, \frac{\sigma^2}{n}\right).$$

Therefore:

$$\frac{\bar{X}-\theta_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1).$$

So,

$$\left(\frac{\bar{X}-\theta_0}{\sigma/\sqrt{n}}\right)^2 \sim \chi^2(1).$$

Thus,

$$\lambda(X) = -\frac{n(\bar{X}-\theta_0)^2}{\sigma^2} \sim -\chi^2(1).$$

(iii) Specify Rejection Region R for Test Size α

To determine the critical value c for test size α, we use the χ^2 distribution. We reject H_0 if:

$$\lambda(X) > c,$$

where c is chosen such that:

$$P(\lambda(X) > c \mid H_0) = \alpha.$$

Since $\lambda(X) \sim -\chi^2(1)$, we need to find c such that:

$$P(-\chi^2(1) > c) = \alpha.$$

This corresponds to:

$$P(\chi^2(1) < -c) = \alpha.$$

Since the $\chi^2(1)$ distribution is non-negative, this condition translates into a comparison with the upper critical value $\chi^2_{1-\alpha}(1)$. Hence,

$$-c = \chi^2_{1-\alpha}(1),$$

$$c = -\chi^2_{1-\alpha}(1).$$

In new variables:

$$Z = \frac{\bar{X}-\theta_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1).$$

The test statistic in standard normal distribution form is:

$$T = \frac{\bar{X}-\theta_0}{\sigma/\sqrt{n}}.$$

The rejection region in terms of Z is:

$$R = \left\{ X : \left| \frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} \right| > z_{1-\alpha/2} \right\},$$

where $z_{1-\alpha/2}$ is the critical value from the standard normal distribution corresponding to the two-tailed test at significance level α.

Reference:

Find below the link to the ChatGpt 3.5/4o history that assisted me during the assignment.
https://chatgpt.com/share/6cb536aa-3e10-46d3-8a1b-8f0c07e3dab3
https://chatgpt.com/share/c7e26eb7-09f3-4d55-98d2-740d56faf765